SEVCON Gen4, Nano, Evo5, GpAC and espAC Fault Codes

Level	LED Flashes	UID	FID	DEC	Type	Display	Message	Description	Recommended Action
1	2	1	0x4481	17537	Warning	F12001	Handbrake Fault	Handbrake is active when direction selected.	Release handbrake
1	5	1	0x4541	17729	Warning	F15001	Fan Fault	No speed feedback from external heatsink fans	Check operation of heatsink fans
1	5	2	0x4542	17730	Warning	F15002	Device test waiting		
1	5	2	0x4542	17730	Warning	F15002	Low Oil	Low Oil	Check oil level
1	5	3	0x4543	17731	Warning	F15003	Hydraulic filter	Hydraulic Filter	Check hydraulic filter
1	5	4	0x4544	17732	Warning	F15004	Pump Current Low	Pump motor is not drawing sufficient current	Check pump motor is connected
1	5	5	0x4545	17733	Warning	F15005	Isolation Fault	Isolation fault detected between logic and power frame	Check isolation between low and high voltage circuits
1	5	6	0x4546	17734	Warning	F15006	No Motor Speed Signal	No speed feedback from motor	Check encoder wiring and speed measurement signal
1	5	7	0x4547	17735	Warning	F15007	Tow Mode	Tow mode has been activated	Disable tow mode if not required
1	5	8	0x4548	17736	Warning	F15008	Steer Sensor Warning	Invalid steer sensor state	Check steer sensor wiring
1	5	9	0x4549	17737	Warning	F15009	Pulsed Enable	Pulsed enable signal not present, unable to enable bridge	Check 1kHz pulsed enable signal is received
1	5	10	0x454A	17738	Warning	F15010	Bridge Enable Delayed	Unable to complete power on checks to allow the bridge to enable	Check motor connected or wait for motor to slow down
1	5	11	0x454B	17739	Warning	F15011	MOSFET s/c tests waiting	MOSFET / IGBT s/c tests at power up are being held off	Check motor connected or wait for motor to slow down
1	5	12	0x454C	17740	Warning	F15012	Electrolyte Low Level	Battery Electrolyte Low Level detected	Check batteries Electrolyte Level
1	5	13	0x454D	17741	Warning	F15013	Electrolyte Cutout Level	Battery Electrolyte Low Level detected and cutbcak	Check batteries Electrolyte Level
1	6	1	0x4581	17793	Warning	F16001	Throttle Fault (Warn)	Warning level throttle fault. Used for RTy	Check throttle wiring and installation.
1	6	2	0x4582	17794	Warning	F16002	Safety Case 1	Throttle appears to be stuck. This fault will clear if throttle starts to work again.	Check throttle wiring and installation.
1	6	3	0x4583	17795	Warning	F16003	Safety Case 2	Throttle appears to be stuck. This fault will latch and can only be cleared by repairing the throttle and recycling power.	Check throttle wiring and installation.
1	6	4	0x4584	17796	Warning	F16004	Analogue Output Over Current (warn)	Contactor driver over current	Ensure contactor doesn't exceed maximum current and check contactor wiring
1	6	5	0x4585	17797	Warning	F16005	Analogue Output Off with Failsafe (warn)	Contactor driver not working	Internal hardware fault
1	6	6	0x4586	17798	Warning	F16006	Analogue Output Over Temperature (warn)	Contactor driver over temperature	Ensure contactor doesn't exceed maximum current and check contactor wiring
1	6	7	0x4587	17799	Warning	F16007	Analogue Output Under Current (warn)	Contactor driver unable to achieve current target in current mode	Ensure contactor driver current target is within range
1	6	8	0x4588	17800	Warning	F16008	Analogue Output Short Circuit (warn)	Contactor driver MOSFET short circuit detected	Internal hardware fault
1	6	9	0x4589	17801	Warning	F16009	Analogue supply (warn)	Analogue supply is >10% out of range	remove short, check output voltage.
1	6	10	0x458A	17802	Warning	F16010	Seat (warning)	seat regen settings applied	Close the seat switch and recycle the direction to remove seat regen settings.
1	7	1	0x45C1	17857	Warning	F17001	BDI Warning	BDI remaining charge (0x2790,1) is less than BDI Warning level (0x2C30,5)	Charge battery
1	7	2	0x45C2	17858	Warning	F17002	BDI Cutout	BDI remaining charge (0x2790,1) is less than BDI Cutout level (0x2C30,4)	Charge battery
1	7	3	0x45C3	17859	Warning	F17003	Low Battery Cut	Battery voltage (0x5100,1) is less than Under Voltage limit (0x2C02,2) for longer than the protection delay (0x2C03,0)	Charge battery
1	7	4	0x45C4	17860	Warning	F17004	High Battery Cut	Battery voltage (0x5100,1) is greater than Over Voltage limit (0x2C01,2) for longer than the protection delay (0x2C03,0)	Charge battery
1	7	5	0x45C5	17861	Warning	F17005	High Capacitor Cut	Capacitor voltage (0x5100,3) is greater than Over Voltage limit (0x2C01,2) for longer than the protection delay (0x2C03,0)	Charge battery
1	7	6	0x45C6	17862	Warning	F17006	Vbat below rated min	Battery voltage (0x5100,1) is less than rated minimum voltage for controller for longer than 1s. NOTE: This fault is sometimes seen at power down.	Charge battery
1	7	7	0x45C7	17863	Warning	F17007	Vbat above rated max	Battery voltage (0x5100,1) is greater than rated maximum voltage for controller for longer than 1s.	Charge battery
1	7	8	0x45C8	17864	Warning	F17008	Vcap above rated max	Capacitor voltage (0x5100,3) is greater than rated maximum voltage for controller for longer than 1s.	Charge battery
1	7	9	0x45C9	17865	Warning	F17009	Motor in low voltage cutback	Motor control (DC link) has entered low voltage cutback region.	Charge battery
1	7	10	0x45CA	17866	Warning	F17010	Motor in high voltage cutback	Motor control (DC link) has entered high voltage cutback region.	Charge battery
1	7	11	0x45CB	17867	Warning	F17011	Mains Under Voltage	Mains Under Voltage	Check Mains Supply
1	7	12	0x45CC	17868	Warning	F17012	Mains Over Voltage	Mains Over Voltage	Check Mains Supply
1	7	13	0x45CD	17869	Warning	F17013	KL15/30 Too Low	The voltage applied to the KL15/30 is below the controller rated minimum	Check KL15/30 supply voltage
1	7	14	0x45CE	17870	Warning	F17014	Vcap Vkey Diff	Vcap is NOT > 7/8 Vkey	Charge battery

Level	LED Flashes	UID	FID	DEC	Type	Display	Message	Description	Recommended Action
1	8	1	0x4601	17921	Warning	F18001	Device too cold	Low heatsink temperature (0x5100,4) has reduced power to motor	Allow controller to warm up to normal operating temperature.
1	8	2	0x4602	17922	Warning	F18002	Device too hot	High heatsink temperature (0x5100,4) has reduced power to motor	Allow controller to cool down to normal operating temperature.
1	8	3	0x4603	17923	Warning	F18003	Motor in thermal cutback	High measured (0x4600,3) or estimated (0x4602,8) motor temperature has reduced power to motor	Allow motor to cool down to normal operating temperature.
1	8	4	0x4604	17924	Warning	F18004	Motor too cold	Low Measured temperature has reached -30deg	Check motor thermistor connection or allow motor to warm up.
1	10	1	0x4681	18049	Warning	F10101	Unit in preoperational	Controller is in pre-operational state	If configured and ready for use, change state to operational.
1	10	2	0x4682	18050	Warning	F10102	IO can't init	Controller has not received all configured RPDOs at power up	Check PDOs on all CANbus nodes are configured correctly and match up.
1	10	3	0x4683	18051	Warning	F10103	RPDO Timeout (warning)	One or more configured RPDOs not received with 3s at start up or 500ms during normal operation.	Check status of all nodes on CANbus. Check PDOs on all CANbus nodes are configured correctly and match up.
1	11	1	0x46C1	18113	Warning	F11101	Encoder Alignment Warning	Encoder is not aligned properly.	Ensure encoder offset is correctly set or re-align encoder
1	11	2	0x46C2	18114	Warning	F11102	SinCos Tracking Warning	SinCos Encoder Min Max Values are heading towards a voltage rail or converging together.	Thermal or mechanical variation is causing the sincos encoder to deviate from the cold factory commissioned values.
1	11	3	0x46C3	18115	Warning	F11103	Fault Ride Through	A fault ride through event has been encountered, but operation is allowed to continue although the system may be derated	none
1	11	4	0x46C4	18116	Warning	F11104	Motor Pull-Out Warning	The open loop V/f induction motor software has detected a pulled-out condition (applied frequency much larger than actual), it will automatically attempt to restart.	Reduce load torque transients and demanded acceleration rates. Check motor parameters (V/f ratio and current limit/allowed torque may be too low) and control gains
1	11	5	0x46C5	18117	Warning	F11105	Stator Resistance Error	Stator resistance thermal compensation has failed due to the estimated value being too different from the configured value	Check the value of Rs configured in the controller is correct for the motor
1	11	6	0x46C6	18118	Warning	F11106	Encoder PLL Deactivated Warning	Encoder PLL has turned off due to poor tracking	Check PLL gains and encoder setup and wiring
1	12	1	0x4701	18177	Warning	F12101	CAN warning	Vehicle is operating in reduced power mode as some CAN messages are not being received (RTy only)	Check status of nodes on CANbus expected to be transmitting data
1	13	1	0x4741	18241	Warning	F13101	Scheduler stack overflow warning	This warning is set if any task has between 10% and 20% of its stack free.	
1	13	2	0x4742	18242	Warning	F13102	Internal supply out of range warning	An internal logic supply rail has gone out of range	
1	14	1	0x4781	18305	Warning	F14101	CANopen anon EMCY level 1	EMCY message received from non-Sevcon node and anonymous EMCY level (0x2830,0) is set to 1.	Check status of non-Sevcon nodes on CANbus
1	14	2	0x4782	18306	Warning	F14102	24V Supply Low		
1	14	3	0x4783	18307	Warning	F14103	24V Supply High		
1	15	1	0x47C1	18369	Warning	F15101	Vehicle Service Required	Vehicle service next due time (0x2850,5) has expired. If supported Service driveability profile (0x2925) will activate.	Service vehicle and reset service hours counter
2	2	1	0x4881	18561	Drive Inhibit	F22001	Seat Fault	Valid direction selected with operator not seated or operator is not seated for a user configurable time in drive.	Must be seated with switches inactive
2	2	2	0x4882	18562	Drive Inhibit	F22002	Two Direction Fault	Both the forward and reverse switches have been active simultaneously for greater than 200 ms.	Check vehicle wiring and reset switches
2	2	3	0x4883	18563	Drive Inhibit	F22003	SRO Fault	FS1 active for user configurable delay (0x2914,2) without a direction selected.	Deselect FS1
2	2	4	0x4884	18564	Drive Inhibit	F22004	Sequence Fault	Any drive switch active at power up.	Deselect all drive switches
2	2	5	0x4885	18565	Drive Inhibit	F22005	FS1 Recycle Fault	FS1 active after a direction change and FS1 recycle function enabled (0x2914,1 bit 1)	Deselect FS1
2	2	6	0x4886	18566	Drive Inhibit	F22006	Inch Fault	Inch switch active along with any drive switch active (excluding inch switches), seat switch indicating operator present or handbrake switch active.	
2	2	7	0x4887	18567	Drive Inhibit	F22007	Overload Fault	Vehicle overloaded	Remove overload condition
2	2	8	0x4888	18568	Drive Inhibit	F22008	Raised and Tilted Fault	Scissor lift platform raised and tilted	Lower platform
2	2	9	0x4889	18569	Drive Inhibit	F22009	Pothole Fault	Scissor lift pothole protection active	Move vehicle out of pot hole.
2	2	10	0x488A	18570	Drive Inhibit	F22010	Traction Inhibit Fault	Traction function inhibited using traction inhibit switch (0x2137)	Deselect traction inhibit.
2	2	11	0x488B	18571	Drive Inhibit	F22011	Illegal Mode Change Fault	Vehicle changed from traction mode to pump mode (or vice versa) when direction selected	Deselect all drive switches
2	2	12	0x488C	18572	Drive Inhibit	F22012	Tilt Sensor Fault	Aichi error code (0x3802,0) set to 0x02	Check tilt sensor
2	2	13	0x488D	18573	Drive Inhibit	F22013	Belly fault	Belly function has activated.	Deselect belly switch
2	2	14	0x488E	18574	Drive Inhibit	F22014	Mom dir fault	Fault with momentary direction selection switch	Release momentary direction switch
2	2	15	0x488F	18575	Drive Inhibit	F22015	Sensorless Startup	Sensorless has failed to start up.	
2	5	1	0x4941	18753	Drive Inhibit	F25001	Motor Overspeed		
2	5	2	0x4942	18754	Drive Inhibit	F25002	PST Fault	An issue has occurred with the PST unit	Check PST unit
2	6	1	0x4981	18817	Drive Inhibit	F26001	Throttle Fault	Throttle value (0x2620,0) is greater than 20% at power up.	Release throttle

Level	LED Flashes	UID	FID	DEC	Type	Display	Message	Description	Recommended Action
2	6	2	0x4982	18818	Drive Inhibit	F26002	E-Brake Wire off	Wire-off detected in electrobrake circuit	Check electrobrake wiring, ensure current flows when energised
2	6	3	0x4983	18819	Drive Inhibit	F26003	Direction Change	Direction is changed and vehicle speed is greated than allowed in 0x2929,2	Neutral switch should be selected to clear the fault
2	7	2	0x49C2	18882	Drive Inhibit	F27002	Entering Cutback	Controller has entered thermal or voltage cutback region	Check for temperature or voltage cutback condition and take appropriate action
2	8	1	0x4A01	18945	Drive Inhibit	F28001	Cutback	Thermal or voltage cutback factors have reduced belowed user defined levels.	Check for temperature or voltage cutback condition and take appropriate action
2	10	1	0x4A81	19073	Drive Inhibit	F20101	RPDO Timeout (drive inhibit)	One or more configured RPDOs not received with 3s at start up or 500ms during normal operation.	Check status of all nodes on CANbus. Check PDOs on all CANbus nodes are configured correctly and match up.
2	12	1	0x4B01	19201	Drive Inhibit	F22101	CAN off bus (drive inhibit)	CANbus off fault condition detected on multinode system. NOTE: This fault was added for Aichi, to replace Very Severe CAN off fault	Check CANbus wiring
2	12	2	0x4B02	19202	Drive Inhibit	F22102	Ren Data	Data missing from CAN (RTy only)	Check connection to CANbus, ensure all devices on bus are communicating.
2	12	3	0x4B03	19203	Drive Inhibit	F22103	IO Data Error	Vehicle application subsystem is not receiving control or status information.	Check CANbus wiring and configuration of J1939 signal mappings.
2	14	1	0x4B81	19329	Drive Inhibit	F24101	CANopen anon EMCY level 2	EMCY message received from non-Sevcon node and anonymous EMCY level (0x2830,0) is set to 2.	Check status of non-Sevcon nodes on CANbus
2	14	3	0x4B83	19331	Drive Inhibit	F24103	Boost over voltage	Boost stage is above maximum operating voltage	
3	1	1	0x4C41	19521	Severe	F31001	Too many slaves	Number of slaves (0x2810,0) set higher than maximum allowed number of slaves	Check 0x2810,0 setting
3	2	1	0x4C81	19585	Severe	F32001	HVIL loop broken	On a product with an High Voltage InterLock feature where the feature is enabled, the loop has been detected as broken.	Check product wiring, cycle KL15/power when fault corrected.
3	4	1	0x4D01	19713	Severe	F34001	Circuit Breaker Open		
3	4	2	0x4D02	19714	Severe	F34002	Circuit Breaker Welded		
3	4	3	0x4D03	19715	Severe	F34003	DC Link Collapsed		
3	4	4	0x4D04	19716	Severe	F34004	Circuit Breaker Timeout	Time to close breaker of GpAC has expired	Cycle power and restart breaker close sequence
3	5	1	0x4D41	19777	Severe	F35001	Motor Isolation Fault	Motor isolation contactor is open circuit	Check isolation contactor and wiring
3	5	2	0x4D42	19778	Severe	F35002	Motor Open Circuit Fault	Motor terminal is open circuit or disconnected from controller	Check motor wiring. Check controller condition
3	5	3	0x4D43	19779	Severe	F35003	Motor stalled	No speed feedback from motor	Check encoder wiring and speed measurement signal
3	7	3	0x4DC3	19907	Severe	F37003	Power Supply Critical	Battery voltage has dropped below critical level	Check controller voltage supply
3	10	1	0x4E81	20097	Severe	F30101	RPDO Timeout (severe)	One or more configured RPDOs not received with 3s at start up or 500ms during normal operation.	Check status of all nodes on CANbus. Check PDOs on all CANbus nodes are configured correctly and match up.
3	11	1	0x4EC1	20161	Severe	F31101	Cant Establish Fld Curr	Controller is unable to control a current in the field winding	Check field is wired to correct terminals
3	11	2	0x4EC2	20162	Severe	F31102	Pulsing Disabled	The motor control protection subsystem unexpectedly disabled the PWM	Check for presence of other faults that may indicate root cause of problem.
3	12	1	0x4F01	20225	Severe	F32101	Unexpected slave state	CANopen slave has changed to unexpected state	Check status of all nodes on CANbus.
3	12	2	0x4F02	20226	Severe	F32102	EMCY send failed	Unable to transmit EMCY message	Internal software fault
3	13	1	0x4F41	20289	Severe	F33101	Internal Fault	Internal software fault	Internal software fault
3	13	2	0x4F42	20290	Severe	F33102	Out of memory	Out of memory	Internal software fault
3	13	3	0x4F43	20291	Severe	F33103	General DSP error	Unknown error raised by motor model code	Internal software fault
3	13	4	0x4F44	20292	Severe	F33104	Timer Failed	Unable to allocate timer	Internal software fault
3	13	5	0x4F45	20293	Severe	F33105	Queue Error	Unable to post message to queue	Internal software fault
3	13	6	0x4F46	20294	Severe	F33106	Scheduler Error	Unable to create task in scheduler	Internal software fault
3	13	7	0x4F47	20295	Severe	F33107	DSP Heartbeat Error	Communication lost between host and DSP processors	Internal hardware fault
3	13	8	0x4F48	20296	Severe	F33108	I/O SS Error	Internal software fault	Internal software fault
3	13	9	0x4F49	20297	Severe	F33109	GIO SS Error	Internal software fault	Internal software fault
3	13	10	0x4F4A	20298	Severe	F33110	LCM SS Error	Internal software fault	Internal software fault
3	13	11	0x4F4B	20299	Severe	F33111	LCP SS Error	Internal software fault	Internal software fault
3	13	12	0x4F4C	20300	Severe	F33112	OBD SS Error	Internal software fault	Internal software fault
3	13	13	0x4F4D	20301	Severe	F33113	VA SS Error	Internal software fault	Internal software fault
3	13	14	0x4F4E	20302	Severe	F33114	DMC SS Error	Internal software fault	Internal software fault
3	13	15	0x4F4F	20303	Severe	F33115	TracApp SS Error	Internal software fault	Internal software fault
3	13	16	0x4F50	20304	Severe	F33116	New Powerframe Detected	New power frame detected.	Recycle keyswitch
3	13	17	0x4F51	20305	Severe	F33117	DSP Not Detected	Communication lost between host and DSP processors	Internal hardware fault
3	13	18	0x4F52	20306	Severe	F33118	DSP Comms Error	Communication lost between host and DSP processors (or IOP and MCP)	Internal hardware fault
3	13	19	0x4F53	20307	Severe	F33119	App Manager SS Error	Internal software fault	Internal software fault
3	13	20	0x4F54	20308	Severe	F33120	Autozero range error	Current sensor auto-zero current out of range	Internal hardware fault

Level	LED Flashes	UID	FID	DEC	Type	Display	Message	Description	Recommended Action
3	13	21	0x4F55	20309	Severe	F33121	DSP parameter error	Communication error between host and DSP processors	Internal software fault
3	13	22	0x4F56	20310	Severe	F33122	Motor in wrong direction	Motor rotation detected as wrong direction. No longer supported	Check motor wiring.
3	13	23	0x4F57	20311	Severe	F33123	Motor stalled	Motor rotation stalled. No longer supported	Check motor wiring.
3	14	1	0x4F81	20353	Severe	F34101	CANopen anon EMCY level 3	EMCY message received from non-Sevcon node and anonymous EMCY level (0x2830,0) is set to 3.	Check status of non-Sevcon nodes on CANbus
4	1	1	0x5041	20545	Very Severe	F41001	Bad NVM Data	EEPROM or flash configuration data corrupted and data can not be recovered.	
4	1	2	0x5042	20546	Very Severe	F41002	VPDO Out of Range	VPDO mapped to non-existent or invalid object	Check all VPDO mappings (0x3000 to 0x3400)
4	1	3	0x5043	20547	Very Severe	F41003	Static Range Error	At least one configuration object is out of range	Set configuration object to valid value. Our of range object can be identified using 0x5621 or Engineering DVT CLI window.
4	1	4	0x5044	20548	Very Severe	F41004	Dynamic Range Error	At least one configuration object is out of dynamic range. This is where one objects range depends on another object.	Check all dynamic range objects. Engineering DVT CLI window indicates type of object which is out of range.
4	1	5	0x5045	20549	Very Severe	F41005	Auto-configuration Fault	Unable to automatically configure I/O and vehicle setup.	Check auto configuration objects (0x5810 and 0x5811)
4	1	6	0x5046	20550	Very Severe	F41006	Voltage autoconfig error	Unable to set battery voltage	Check auxillary drives support low voltage configurations
4	2	1	0x5081	20609	Very Severe	F42001	Invalid Steer Switches	Steering switches are in an invalid state	Check steering switches and wiring
4	4	1	0x5101	20737	Very Severe	F44001	Line Contactor o/c	Line contactor did not close when coil is energized.	Check line contactor and wiring
4	4	2	0x5102	20738	Very Severe	F44002	Line Contactor welded	Line contactor closed when coil is denergized.	Check line contactor and wiring
4	5	1	0x5141	20801	Very Severe	F45001	Beltloader Fault	Unable to change between traction and pump motors on beltloader.	Check change over contactors and motor wiring.
4	5	2	0x5142	20802	Very Severe	F45002	Ren Signal	Fault signalled by RTy vehicle network	Check peripheral RTy devices
4	5	3	0x5143	20803	Very Severe	F45003	VERLOG	VERLOG signal failure	Check peripheral RTy devices
4	6	1	0x5181	20865	Very Severe	F46001	Digital Input Wire Off	Digital input wire-off	Check wiring
4	6	2	0x5182	20866	Very Severe	F46002	Analogue Input Wire Off	Analogue input outside of allowed range (0x46cX)	Check wiring
4	6	3	0x5183	20867	Very Severe	F46003	Analogue Output Over Current	Contactor driver over current	Ensure contactor doesn't exceed maximum current and check contactor wiring
4	6	4	0x5184	20868	Very Severe	F46004	Analogue Output On with No Failsafe	Internal hardware failsafe circuitry not working	Internal hardware fault
4	6	5	0x5185	20869	Very Severe	F46005	Analogue Output Off with Failsafe	Contactor driver not working	Internal hardware fault
4	6	6	0x5186	20870	Very Severe	F46006	Analogue Output Over Temperature	Contactor driver over temperature	Ensure contactor doesn't exceed maximum current and check contactor wiring
4	6	7	0x5187	20871	Very Severe	F46007	Analogue Output Under Current	Contactor driver unable to achieve current target in current mode	Ensure contactor driver current target is within range
4	6	8	0x5188	20872	Very Severe	F46008	Analogue Output Short Circuit	Contactor driver MOSFET short circuit detected	Internal hardware fault
4	7	1	0x51C1	20929	Very Severe	F47001	Power Supply Interrupt	Not used	
4	7	2	0x51C2	20930	Very Severe	F47002	Capacitor Precharge Failure	Capacitor voltage (0x5100,3) did not rise above 5V at power up	Check power wiring
4	7	3	0x51C3	20931	Very Severe	F47003	KL15/30 Too High	The voltage applied to the KL15/30 is above the controller rated maximum	Check KL15/30 supply voltage
4	8	1	0x5201	20993	Very Severe	F48001	Heatsink overtemp	Controller heat sink has reached critical high temperature, and has shut down.	Allow controller to cool down to normal operating temperature.
4	11	1	0x52C1	21185	Very Severe	F41101	Encoder Fault	Encoder input wire-off is detected.	Check encoder wiring
4	11	2	0x52C2	21186	Very Severe	F41102	Motor Overcurrent Fault	Motor current exceeded controller rated maximum	Check motor configuration and wiring
4	11	3	0x52C3	21187	Very Severe	F41103	Current Control Fault	Motor controller unable to maintain control of motor currents	Check motor configuration. Ensure motor speed is not too high.
4	11	4	0x52C4	21188	Very Severe	F41104	Motor Overspeed Fault	Motor control tripped due to motor overspeed	Check motor configuration. Ensure motor speed is not too high.
4	11	5	0x52C5	21189	Very Severe	F41105	Encoder Alignment Severe	Encoder is not aligned properly.	Ensure encoder offset is correctly set or re-align encoder
4	11	6	0x52C6	21190	Very Severe	F41106	di/dt fault (suspected s/c)	Large rate of change of current detected multiple times. Suspected MOSFET, motor or wiring short circuit.	Check motor wiring - especially for motor terminals shorted to B-. Check controller condition.
4	11	7	0x52C7	21191	Very Severe	F41107	Vcap Overvotlage (software overvoltage)	Measured capacitor voltage has exceeded controller maximum.	Check battery condition and wiring
4	12	1	0x5301	21249	Very Severe	F42101	CANBUS Fault	CANbus fault condition detected on multinode system.	Check CANbus wiring
4	12	2	0x5302	21250	Very Severe	F42102	Bootup not received	CANopen slave has not transmitted boot up message at power up	Check status of all nodes on CANbus.
4	12	3	0x5303	21251	Very Severe	F42103	LPRX queue overrun	CANbus fault condition detected on multinode system.	Check CANbus wiring
4	12	4	0x5304	21252	Very Severe	F42104	LPTX queue overrun	CANbus fault condition detected on multinode system.	Check CANbus wiring
4	12	5	0x5305	21253	Very Severe	F42105	HPRX queue overrun	CANbus fault condition detected on multinode system.	Check CANbus wiring
4	12	6	0x5306	21254	Very Severe	F42106	HPTX queue overrun	CANbus fault condition detected on multinode system.	Check CANbus wiring
4	12	7	0x5307	21255	Very Severe	F42107	CAN overrun	CANbus fault condition detected on multinode system.	Check CANbus wiring
4	12	8	0x5308	21256	Very Severe	F42108	CAN off bus	CANbus fault condition detected on multinode system.	Check CANbus wiring
4	12	9	0x5309	21257	Very Severe	F42109	Nodeguarding Failed	Not used	

Level	LED Flashes	UID	FID	DEC	Type	Display	Message	Description	Recommended Action
4	12	10	0x530A	21258	Very Severe	F42110	Short PDO received	Received RPDO doesn't contains enough bytes	Check PDOs on all CANbus nodes are configured correctly and match up.
4	12	11	0x530B	21259	Very Severe	F42111	CANopen Heartbeat Failed	Heartbeat not received within configured time out (0x1016)	Check status of all nodes on CANbus.
4	12	12	0x530C	21260	Very Severe	F42112	CANopen slave in wrong state	CANopen slave has changed to unexpected state	Check status of all nodes on CANbus.
4	12	13	0x530D	21261	Very Severe	F42113	CAN ESTAT set	Internal CANbus fault	Internal software fault
4	12	14	0x530E	21262	Very Severe	F42114	SDO HDL Error	Internal CANbus fault	Internal software fault
4	12	15	0x530F	21263	Very Severe	F42115	SDO Timeout Error	Internal CANbus fault	Internal software fault
4	12	16	0x5310	21264	Very Severe	F42116	SDO Abort Error	Internal CANbus fault	Internal software fault
4	12	17	0x5311	21265	Very Severe	F42117	SDO State Error	Internal CANbus fault	Internal software fault
4	12	18	0x5312	21266	Very Severe	F42118	SDO Toggle Error	Internal CANbus fault	Internal software fault
4	12	19	0x5313	21267	Very Severe	F42119	SDO Rec Error	Internal CANbus fault	Internal software fault
4	12	20	0x5314	21268	Very Severe	F42120	SDO Len Error	Internal CANbus fault	Internal software fault
4	12	21	0x5315	21269	Very Severe	F42121	SDO Send Error	Internal CANbus fault	Internal software fault
4	12	22	0x5316	21270	Very Severe	F42122	SDO unknown event	Internal CANbus fault	Internal software fault
4	12	23	0x5317	21271	Very Severe	F42123	SDO Bad SRC	Internal CANbus fault	Internal software fault
4	12	24	0x5318	21272	Very Severe	F42124	SDO bad error number	Internal CANbus fault	Internal software fault
4	12	25	0x5319	21273	Very Severe	F42125	Motor slave in wrong state	Motor slave in wrong state	Check status of all nodes on CANbus controlling motor slaves. Check local motor slaves on master. Ensure configuration is correct.
4	12	26	0x531A	21274	Very Severe	F42126	Ren Protocol	CAN device on RTy not responding	Check connection to CANbus, ensure all devices on bus are communicating.
4	13	1	0x5341	21313	Very Severe	F43101	Invalid DSP Protocol	DSP reports invalid protocol version on dual processor platform	Internal software fault
4	13	2	0x5342	21314	Very Severe	F43102	OSC Watchdog Fault	Internal hardware fault	Internal hardware fault
4	13	3	0x5343	21315	Very Severe	F43103	Fault List Overflow	Attempting to set too many faults.	Internal software fault
4	13	4	0x5344	21316	Very Severe	F43104	DSP SPI Comms Fault	Communication error between host and DSP processors (or gate drive and MCP)	Internal hardware fault
4	13	5	0x5345	21317	Very Severe	F43105	Scheduler stack overflow fault	Less than 10% of the stack is free on one of the RTOS tasks.	Internal hardware fault
4	13	6	0x5346	21318	Very Severe	F43106	Internal supply out of range fault	An internal logic supply rail has gone out of range	
4	14	1	0x5381	21377	Very Severe	F44101	CANopen anon EMCY level 4	EMCY message received from non-Sevcon node and anonymous EMCY level (0x2830,0) is set to 4.	Check status of non-Sevcon nodes on CANbus
5	1	1	0x5441	21569	Return to Base	F51001	Incompatible hardware version	Detected controller hardware version incompatible with software	Check correct software is programmed into controller. Reprogram if necessary
5	1	2	0x5442	21570	Return to Base	F51002	Calibration Fault	Calibration settings in controller are out of range	Controller requires recalibration in production
5	3	1	0x54C1	21697	Return to Base	F53001	PFOvervoltage	Voltage on B+ terminal exceeds rated maximum for controller	Check battery condition and wiring
5	3	2	0x54C2	21698	Return to Base	F53002	Powerframe Fault	Motor current exceeded controller rated maximum	Check motor configuration and wiring
5	3	3	0x54C3	21699	Return to Base	F53003	MOSFET s/c M1>B+	MOSFET / IGBT s/c detection on M1 top devices	Check motor wiring. Check controller condition
5	3	4	0x54C4	21700	Return to Base	F53004	MOSFET s/c M1>B-	MOSFET / IGBT s/c detection on M1 bottom devices	Check motor wiring. Check controller condition
5	3	5	0x54C5	21701	Return to Base	F53005	MOSFET s/c M2>B+	MOSFET / IGBT s/c detection on M2 top devices	Check motor wiring. Check controller condition
5	3	6	0x54C6	21702	Return to Base	F53006	MOSFET s/c M2>B-	MOSFET / IGBT s/c detection on M2 bottom devices	Check motor wiring. Check controller condition
5	3	7	0x54C7	21703	Return to Base	F53007	MOSFET s/c M3>B+	MOSFET / IGBT s/c detection on M3 top devices	Check motor wiring. Check controller condition
5	3	8	0x54C8	21704	Return to Base	F53008	MOSFET s/c M3>B-	MOSFET / IGBT s/c detection on M3 bottom devices	Check motor wiring. Check controller condition
5	3	9	0x54C9	21705	Return to Base	F53009	MOSFET s/c checks incomplete	Unable to complete MOSFET / IGBT s/c tests at power up	Internal software fault
5	3	10	0x54CA	21706	Return to Base	F53010	Pump Mosfet S/C	MOSFET / IGBT s/c detection Pump Mosfet Devices	Check motor wiring. Check controller condition
5	3	10	0x54CA	21706	Return to Base	F53010	IGBT M1 Low Driver Fail	IGBT driver failure	Check status of IGBT
5	3	11	0x54CB	21707	Return to Base	F53011	IGBT M1 High Driver Fail	IGBT driver failure	Check status of IGBT
5	3	12	0x54CC	21708	Return to Base	F53012	IGBT M2 Low Driver Fail	IGBT driver failure	Check status of IGBT
5	3	13	0x54CD	21709	Return to Base	F53013	IGBT M2 High Driver Fail	IGBT driver failure	Check status of IGBT
5	3	14	0x54CE	21710	Return to Base	F53014	IGBT M3 Low Driver Fail	IGBT driver failure	Check status of IGBT
5	3	15	0x54CF	21711	Return to Base	F53015	IGBT M3 High Driver Fail	IGBT driver failure	Check status of IGBT
5	13	1	0x5741	22337	Return to Base	F53101	Invalid Powerframe Rating	Unable to identify hardware	Internal hardware fault
5	14	1	0x5781	22401	Return to Base	F54101	CANopen anon EMCY level 5	EMCY message received from non-Sevcon node and anonymous EMCY level (0x2830,0) is set to 5.	Check status of non-Sevcon nodes on CANbus